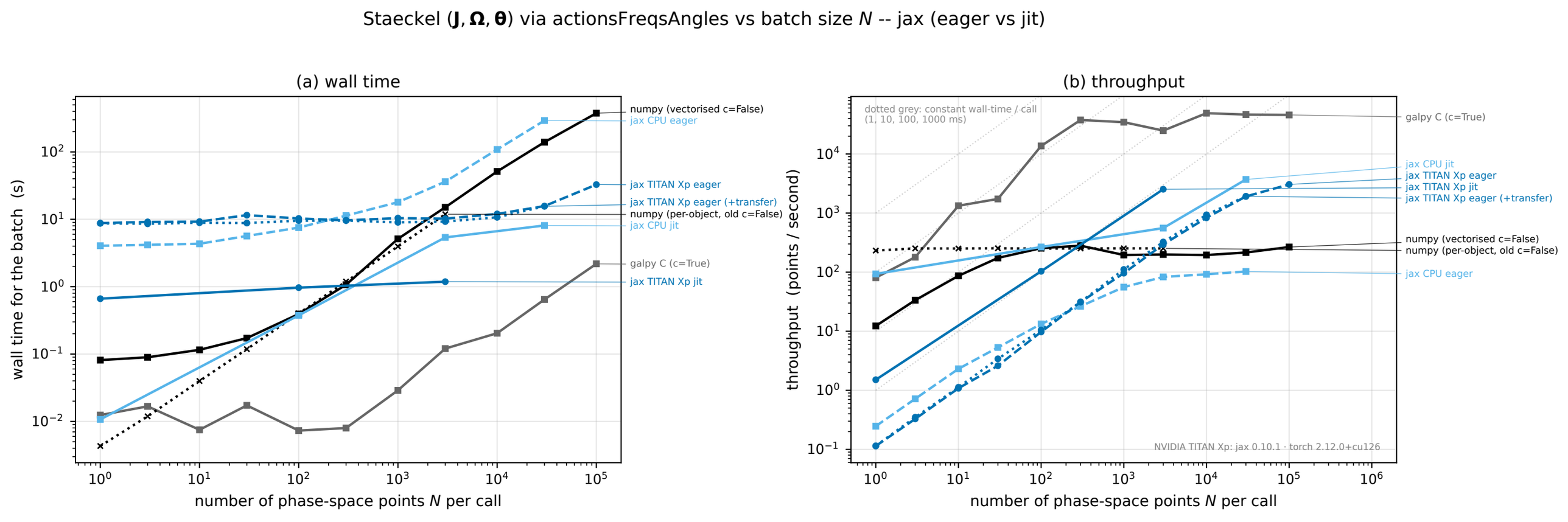
Staeckel ($\mathbf{J}, \mathbf{\Omega}, \boldsymbol{\theta}$) via actionsFreqsAngles vs batch size N -- jax (eager vs jit)
(a) wall time
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised c=False)
jax CPU eager
jax TITAN Xp eager
jax TITAN Xp eager (+transfer)
numpy (per-object, old c=False)
jax CPU jit
galpy C (c=True)
jax TITAN Xp jit
(b) throughput
throughput (points / second)
number of phase-space points N per call
dotted grey: constant wall-time / call
(1, 10, 100, 1000 ms)
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
galpy C (c=True)
jax CPU jit
jax TITAN Xp eager
jax TITAN Xp jit
jax TITAN Xp eager (+transfer)
numpy (vectorised c=False)
numpy (per-object, old c=False)
jax CPU eager